

Optimizing Learning in University: Three Evidence-Based Study Strategies

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Introduction

As a university student, you spend many hours studying. Yet, many students are surprised and disappointed by their results. A common reason is that they rely on strategies that feel effective but are not: rereading a textbook chapter, for example, creates a sense of familiarity that is easily mistaken for real knowledge. Cognitive research has identified study techniques that produce more durable learning, yet students tend to underuse them. Self-report studies show that students favour rereading and highlighting over the methods that research supports (Miyatsu et al., 2018). Some reviews have responded by suggesting how students might improve the strategies they already prefer (Miyatsu et al., 2018) a more direct (one might even say, reasonable) route, however, is to adopt the strategies with the strongest evidence. This essay therefore asks how university students can optimize their learning, and answers by discussing three evidence-based strategies: pretesting, retrieval practice, and spaced practice. Effective studying is not simply a matter of effort, but of method.

Pretesting

The first strategy is pretesting: attempting to answer questions about material before studying it, even if the answers to the questions must be guessed. Although this seems counterintuitive, research indicates that answering questions about a topic before reading about it improves later recall (Putnam et al., 2016). The benefit arises because the attempt activates any prior knowledge and prepares the reader to connect new information with what is already known. Even incorrect guesses are useful, because they make the reader notice and remember the correct information when it appears. Prequizzing is becoming easier to practice. (Putnam et al., 2016) suggests that before a studying session (like a reading session or a lecture), a student can write down a few questions about the topic and attempt to answer them, then return to those questions afterwards to check and correct the answers. However, many textbooks also include postreading quizzed. They can also be easily repurposed for pretesting. Thirdly, the rise of AI gives us new opportunities. A chatbot can create a quiz for you after reading your lecture slides or textbook chapter. It is even possible to prompt it to follow your precise specified needs such as difficulty level. The main obstacle is psychological rather than practical: guessing at material one has not yet studied feels wrong, and students generally prefer to read first. Recognising that pretesting is a low-cost way of preparing the memory, rather than a test of current knowledge, helps to overcome this hesitation.

Retrieval Practice

The second strategy is retrieval practice: recalling information from memory rather than merely reviewing it. A well-known form is the read–recite–review method, in which a student reads a passage, recites the main ideas from memory, and then reviews the passage to check what was missed (Putnam et al., 2016). Other implementations include answering one’s own questions and using flashcards, provided that the answer is produced from memory before it is checked (Miyatsu et al., 2018). In practice, this means that a student might read a section, close the book, write down everything that can be remembered, and only then check the text. Practice testing is among the most effective techniques available, because retrieving information strengthens the memory for it and simultaneously reveals gaps that require further study (Putnam et al., 2016). The obstacle is that retrieval practice feels harder and slower than rereading, while rereading produces a misleading impression of knowing. Students should therefore trust the evidence rather than their feeling of fluency, and should always check their answers and revisit material they could not recall. Applied consistently, retrieval practice turns studying into an active process in which each attempt to remember strengthens later performance.

Spaced Practice

The third strategy is spaced practice: distributing study over time instead of concentrating it into a single session. Cramming may produce acceptable performance in the short term, but spaced study leads to far more durable learning, and it requires no more total time (Putnam et al., 2016). Spacing is also central to the recommendations for the popular strategies that students already use, where repetitions distributed across days improve retention (Miyatsu et al., 2018). To implement it, a student can study each subject a little on most days, revisit earlier topics in later sessions, and plan these sessions in advance. The main obstacles are habit and time pressure: cramming feels productive because it produces rapid gains, and it is the default response when deadlines approach. Scheduling regular short sessions, and accepting that learning may feel slower at first, makes spacing easier to sustain over a semester. Spacing and retrieval practice work well together: each spaced session can begin with a short self-test of the material covered previously. Short, regular sessions are also less stressful than last-minute cramming.

Conclusion

University students can optimize their learning by replacing familiar but passive habits with strategies that actively engage memory and distribute effort over time. Pretesting prepares the memory before learning begins; retrieval practice, especially in the form of read–recite–review, strengthens memory by requiring information to be recalled; and spaced

practice ensures that learning is consolidated over days rather than hours. Together, these techniques demand more effort than rereading or highlighting, but they produce the durable learning that university study requires. Improving the strategies students already use may offer some benefit, but the greater gain lies in adopting the strategies that the evidence supports.

References

Bibliography

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Putnam, A. L., Sungkhasettee, V. W., & Roediger, L., Henry L. (2016). Optimizing Learning in College: Tips From Cognitive Psychology. *Perspectives on Psychological Science*, 11(5), 652–660. <https://doi.org/10.1177/1745691616645770>

AI disclosure statement

An AI assistant was used extensively in the preparation of this essay, and this is disclosed in full. The assistant was opencode, running a DeepSeek language model. It drafted the initial text of the essay on the basis of the two assigned articles and the assignment brief, and it was also used to plan the structure of the essay and to discuss which study strategies to include. Every prompt sent to the assistant, and every reply it gave, are recorded in the project's public repository, together with a word-level attribution of the final text that shows which passages originated with the assistant and which were written or rewritten by the author. All sources cited in the essay were checked against the original articles, and the author reviewed and edited the final text and takes responsibility for its content.

Reflection

Writing this essay was more instructive than I expected. The two articles described study strategies that I have rarely used, and one in particular stood out: the read–recite–review method. Reading about it, I realised that I had planned to recite and review the articles after reading them, but never did, and I am now fairly sure that this is why I remember so little of their detail. The exercise also made me notice how often I choose study methods because they feel productive rather than because they work. In future, I want to try retrieval practice and spacing deliberately, and to pay attention to whether they feel harder at first. The writing itself was manageable, but I found it difficult to keep the text objective and to decide how much detail each strategy needed within the word limit.